

ARCHITECTURE

BobaLink Browser Extension — Architecture

Revision: 1 **Last modified:** 2026-06-10T23:55:00Z **Scope:** BobaLink (extension/) — WXT + TypeScript Manifest-V3 cross-browser extension that detects magnet links and .torrent URLs on supported pages and forwards them to the Boba merge service on port 7187. **Authority:** the real source tree under extension/src/; docs/browser_extension/Status.md (Rev 7).

Anti-bluff (§11.4.6). Every component, edge, and message type below traces to a real module under extension/src/ (verified by reading the source). No invented components. Where a name is load-bearing it is given exactly as it appears in the code (isValidScanResult, MENU_SEND_GROUP, addMagnets, readSessionPassphrase, POST /api/v1/download, ...).

1. Data-flow diagram

The diagram renders on GitHub. In the exported HTML it remains a fenced code block (per scope note) — the textual edges are the load-bearing record.

```
flowchart TD
    subgraph PAGE["Web page (content-script context – matches from SITE_SELECTORS)"]
        DOM["Page DOM<br/>(magnet: links, .torrent URLs, bare text)"]
        ENTRY_C["entrypoints/content.ts<br/>(WXT thin wrapper · run_at document_idle)"]
        CONTENT["content/index.ts<br/>initContentScript()"]
        ORCH["scanner/orchestrator.ts<br/>ScannerOrchestrator (cross-scanner dedup by stable id)"]
    end
```

```

        SCAN["scanner/{link-scanner,text-scanner}.ts<br/>+ base.ts
· site-db.ts"]
        PARSE["parser/{magnet,torrent-file,bencode}.ts<br/>(SHA-1
infohash)"]
        HL["content/highlight.ts<br/>HighlightManager (DOM
marking, toggleable)"]
        end

        subgraph SW["Background service worker (MV3)"]
            ENTRY_B["entrypoints/background.ts<br/>(WXT thin
wrapper)"]
            BG["background/index.ts<br/>handleMessage() router +
initBackground()"]
            GUARD["isValidScanResult()<br/>(trust boundary shape-
guard)"]
            STATE["tabResults: Map<tabId, PageScanResult><br/>+
action badge + notifications"]
            TG["tabgroups/index.ts<br/>batchGroupTorrents /
dispatchGroupBatch"]
            CLIENT["api/boba-client.ts<br/>BobaClient (token-bucket +
retry)"]
            QUEUE["api/queue.ts<br/>OfflineQueue (FIFO, persisted,
dead-letter)"]
            HEALTH["api/health.ts<br/>probeHealth()"]
        end

        subgraph CRED["Credentials (§11.4.10)"]
            CRYPTO["shared/crypto.ts<br/>AES-256-GCM decrypt() ·
PBKDF2"]
            SSTORE["chrome.storage.session<br/>session passphrase (in-
memory only)"]
        end

        subgraph UI["Extension UI pages"]
            POPUP["popup/popup.ts<br/>detected list + Send / Send-
All"]
            OPTIONS["options/options.ts<br/>servers, config, token
encrypt"]
            LSTORE["chrome.storage.local<br/>(shared/storage.ts)"]
        end

        BOBA["Boba merge service :7187 (FastAPI)<br/>POST /api/v1/
download · GET /health · GET /api/v1/auth/status"]

        DOM --> ENTRY_C --> CONTENT --> ORCH
        ORCH --> SCAN --> PARSE
        ORCH -->|scan-completed event| CONTENT
        ORCH -->|torrent-detected event| HL --> DOM
        CONTENT -->|chrome.runtime.sendMessage 'scan-result'| BG
        BG --> GUARD -->|valid| STATE

        POPUP -->|'get-detected'| BG

```

```

BG -->|"{ result }"| POPUP
POPUP -->|'send-torrent' ids| BG
BG -->|MENU_SEND_GROUP| TG --> BG
BG -->|addMagnet / addMagnets| CLIENT
CLIENT -->|"POST /api/v1/download {result_id, download_urls}"|
BOBA
CLIENT -->|send failure → enqueueFailed| QUEUE
QUEUE -->|"processQueue(processQueueItem) retry"| CLIENT
BG -->|'health-check'| HEALTH --> BOBA

CLIENT -->|BobaClient.create decrypts token| CRYPTO
BG -->|readSessionPassphrase| SSTORE --> CRYPTO

OPTIONS -->|'get-config' / 'set-config'| BG
OPTIONS --> LSTORE
BG --> LSTORE

BG -->|"chrome.tabs.sendMessage 'scan-now' / 'highlight-
toggle'"| CONTENT

```

Edge legend. Solid = primary request/response path; dotted = failure/retry path. Quoted labels are real MessageType strings or method names from the code.

Narrative trace (page → backend)

1. **Page DOM → content script.** The WXT content entrypoint (entrypoints/content.ts, run_at: document_idle, matches derived from SITE_SELECTORS) calls initContentScript() (content/index.ts), which builds a ScannerOrchestrator.
2. **Orchestrator → scanners → parsers.** The orchestrator runs LinkScanner + TextScanner over the DOM, deduping by the base.ts stable id; the parsers (magnet.ts, torrent-file.ts, bencode.ts) extract the SHA-1 infohash. On each scan-completed event the content script sends a typed scan-result envelope up via chrome.runtime.sendMessage. The HighlightManager marks detected links in the DOM (toggleable).
3. **Background router.** background/index.ts handleMessage() receives scan-result, passes it through isValidScanResult() (the trust-boundary shape-guard — a non-array items is rejected so a hostile page cannot poison the set), and stores it in tabResults keyed by tab id, updating the action badge.
4. **Send.** On send-torrent (or MENU_SEND_GROUP), the background builds a BobaClient, decrypting the configured token, and calls addMagnet / addMagnets, which POST /api/v1/download to :7187 with {result_id, download_urls:[...]}.

5. **Failure → offline queue → retry.** Each failed send is enqueueFailed-ed into the persisted OfflineQueue; processQueue(processQueueItem) re-attempts queued items through a fresh client, dead-lettering after maxRetries.
 6. **Token decrypt.** Building the client (BobaClient.create) decrypts the AES-256-GCM encryptedBobaApiToken bundle via shared/crypto.decrypt, keyed by the session passphrase read from chrome.storage.session (readSessionPassphrase). Default-open when locked — the ciphertext is never sent.
 7. **Popup / options / health.** The popup queries get-detected and dispatches send-torrent; the options page round-trips config via get-config / set-config; the background's health-check and a periodic alarm probe GET /health via probeHealth.
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2. Module responsibilities

Module (extension/src/...)	Responsibility
parser/bencode.ts	Bencode decoder for .torrent file bodies.
parser/magnet.ts	Parse magnet: URIs; sanitizeDisplayName (ReDoS-hardened linear tag-strip).
parser/torrent-file.ts	Parse .torrent files; compute the SHA-1 infohash.
scanner/base.ts	BaseScanner contract + computeStableId (infohash → magnet URI → file URL → name).
scanner/link-scanner.ts	Detect torrents in <a href> links.
scanner/text-scanner.ts	Detect bare-text magnets / .torrent URLs in page text.
scanner/orchestrator.ts	Run all scanners, cross-scanner dedup by stable id, MutationObserver re-scan; emit scan-completed / torrent-detected.
scanner/site-db.ts	Curated per-site selector table (single source of truth).
content/index.ts	initContentScript() — orchestrator + highlight +

Module (extension/src/...)	Responsibility
	content↔background messaging; sends scan-result.
content/highlight.ts	HighlightManager — DOM marking of detected links, runtime-toggleable.
background/index.ts	MV3 message-hub: handleMessage router, context menus, keyboard commands, alarms, badge, notifications, send + enqueue; isValidScanResult trust-guard; MENU_SEND_GROUP.
popup/popup.ts	initPopup(doc) — detected-torrent list, per-row Send + Send-All, status indicator; talks to background ONLY via messages; safe DOM (no innerHTML).
options/options.ts	Servers/config UI, token encryption, WAI-ARIA tablist (Arrow/Home/End keyboard nav).
api/boba-client.ts	BobaClient — the one network client to :7187; addMagnet/addMagnets (POST /api/v1/download), health, authStatus; token-bucket rate-limit + exponential-backoff retry; BobaClient.create decrypt-and-send.
api/queue.ts	OfflineQueue — persisted FIFO retry queue; priority order, FIFO-evict at maxSize, retry-state, dead-letter; SEND injected as a QueueSender.
api/health.ts	probeHealth() — liveness/latency/version probe of a server.
tabgroups/index.ts	batchGroupTorrents / dispatchGroupBatch — dedupe a tab group, one batched addMagnets POST.
shared/crypto.ts	

Module (extension/src/...)	Responsibility
	AES-256-GCM encrypt/decrypt, PBKDF2 key derivation, EncryptedBundle, isEncrypted.
shared/storage.ts	storageGet/storageSet over chrome.storage.local.
shared/events.ts	TypedEventEmitter + EventMap (in-process domain events; scanner↔content).
shared/logger.ts	createLogger/initLogger (token/passphrase never logged).
shared/constants.ts	SITE_SELECTORS, STORAGE_KEYS, BADGE_COLORS, BOBA_TOKEN_HEADERS, RATE_LIMIT, RETRY_CONFIG, REQUEST_TIMEOUTS, EXT, ENCRYPTION, DEBOUNCE_DELAYS.
shared/utils.ts	TokenBucket, debounce, sleep, truncate, yieldToBrowser.
shared/errors.ts	NetworkError, ServerError, StorageError.
types/{api,config,torrent}.ts	MessageType / ExtensionMessage / PageScanResult / DetectedTorrent / ExtensionConfig / ServerConfig.

3. WXT entrypoint / thin-wrapper layout

WXT generates the MV3 manifest and bundles the worker/scripts from the entrypoints/ directory. Each entrypoint is a **thin wrapper** that calls into a fully unit-testable logic module, so the test corpus imports the logic modules directly while the entrypoints own the real browser lifecycle.

Entrypoint	Wraps	Manifest role
entrypoints/background.ts	initBackground() (background/index.ts)	background.service_worker. Idempotent — safe even if the logic module auto-registers on import.

Entrypoint	Wraps	Manifest role
entrypoints/ content.ts	initContentScript() (content/index.ts)	content_scripts with matches derived from SITE_SELECTORS (no <all_urls>), run_at: document_idle, all_frames: false. SINGLE driver of the content logic (the logic module's self-init was removed to avoid double-registering the message listener).
entrypoints/ popup/ index.html	loads popup/ popup.ts	action.default_popup.
entrypoints/ options/ index.html	loads options/ options.ts	options_ui / options page.

Build output: loadable .output/chrome-mv3/ + .output/firefox-mv2/, plus per-store zips bobalink-1.0.0-{chrome,firefox,sources}.zip (produced by extension/ci-ext.sh, §11.4.38 artifact-verified).

4. Message-type catalog

MessageType is the closed union in types/api.ts. The catalog below lists the messages actually routed by the two handleMessage switches (background in background/index.ts, content in content/index.ts) and their direction.

Sender → Background (routed by background/index.ts)

Type	Sender	Background action
scan-result	content script	Trust-guard via isValidScanResult; store in tabResults; update badge.
get-detected	popup	Return the tab's stored PageScanResult ({ result }).
send-torrent	popup	sendTorrents(tabId, ids) → BobaClient.addMagnet; enqueue failures.

Type	Sender	Background action
health-check	popup / options	probeHealth every configured server; return per-server status.
queue-status	popup / options	Return OfflineQueue size + items.
queue-process	popup / options	offlineQueue.processQueue(processQueueItem) (retry drain).
get-config	popup / options	Return the persisted ExtensionConfig.
set-config	options	Persist config; re-init the logger.
scan-page	popup / command	Forward scan-now to the tab's content script.
open-dashboard	popup	Open the active server URL in a new tab.

Background → Content (chrome.tabs.sendMessage, routed by content/index.ts)

Type	Content action
scan-now	Re-scan the page (rescan() → orchestrator.scanNow); re-sends scan-result.
highlight-toggle	Toggle (or set) HighlightManager DOM highlighting.
get-detected	Return the orchestrator's detected set.
toggle-selection	Flip a detected torrent's selected flag.
get-scan-status	Report scanning/scanned state + count.

The unused-by-router members of the MessageType union (send-result, health-result, get-auth-state, authenticate, show-notification, update-badge, torrent-detected, selection-change) are declared response / reserved variants; the load-bearing routed set is the two tables above.

Sources verified 2026-06-10

- `extension/src/background/index.ts` — message router, `isValidScanResult`, `MENU_SEND_GROUP`, `alarms`, `readSessionPassphrase`, `clientFor`
- `extension/src/content/index.ts` — `initContentScript`, `scan-result send`, content message switch
- `extension/src/api/boba-client.ts` — `POST /api/v1/download`, `addMagnet(s)`, `BobaClient.create decrypt-and-send`
- `extension/src/api/queue.ts` — `OfflineQueue`, injected `QueueSender`
- `extension/src/scanner/orchestrator.ts` — cross-scanner dedup
- `extension/src/shared/{crypto,events,constants,storage}.ts`
- `extension/src/entrypoints/{background.ts,content.ts}` — thin-wrapper layout
- `extension/src/types/api.ts` — `MessageType` union